JALPAIGURI GOVERNMENT ENGINEERING COLLEGE

[A GOVERNMENT AUTONOMOUS COLLEGE]

JGEC/B.TECH/ CSE/PCC-CS302/ 2025-26

2026

Data Structure and Algorithms

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions 5x2=10

1. Find the maximum number of nodes in a binary tree of level k, where k >= 1.(Height=Maximum level+1) 2
2. What is load factor? 2
3. Which of the following is the best and worst time for an algorithm? 2
   a)O(n) b)O(log_2n) c)O(2^n) d) O($nlog_2n$)
4. Explain the advantages of binary search over linear search 2
5. State two applications of stack data structure 2

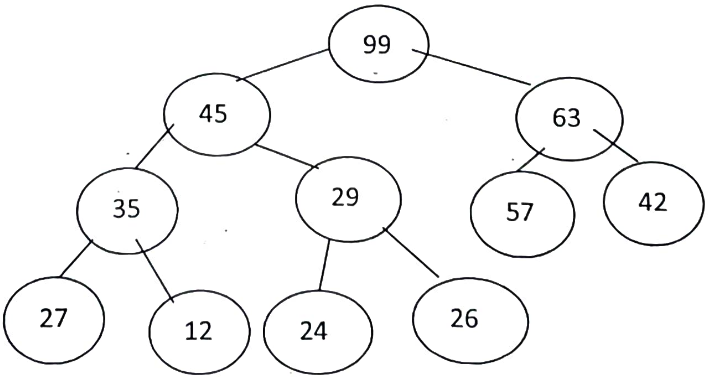
GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions 4x15 = 60

6. i) Show the steps of insertion of the following items in an empty binary search tree 5
   60, 25,75,66,15,50,33,44.

   ii) Now delete the following items from the main tree. (Each value should be deleted from the main tree separately). a) 44 b) 25 c) 75 5

   iii) Delete node 99 from the following max heap tree. Show the steps 5



5*7 9+5*7 -6^2

7. i) Construct a binary tree from pre and in order traversal. 7
   Pre-order- A B D I E J C F G K
   In-order - D I B E J A F C K G

   ii) A two-dimensional array TABLE [5] [5] is stored with base address 351. What is the address of TABLE [2] [4] in terms of Row major and Column major? 6

   iii) What is doubly circular linked list? 2

8. i) What is the limitation of a simple queue? How can you overcome this limitation? 3

   ii) Convert the infix expression 9 +5* 7-6^2+15/3 into equivalent postfix expression using stack data structure and evaluate the postfix expression. Clearly show the state of the stack. (^ has the highest priority among these operators. Use space or comma to distinguish different number for convenience) 7+5

9. i) Write an algorithm to create a single linked list. What are the advantages of using linked list over array 7+2

i) Write an algorithm to insert an element in a circular queue 6

10. i) The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function h(k) = k mod 10 and linear probing. Show each step and the resultant hash table? 7

ii) What are the advantages of AVL tree over binary search tree?Explain parallel edge and loop in a graph with an example 3+3

iii) What is complete binary tree? 2

11. Write short notes on any three of the following topics 5+5+5

a)B tree

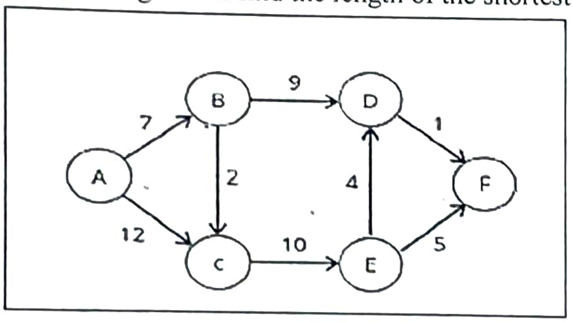
b) Collision Resolution techniques in Hashing

c) Representation of Graph

d)DFS

e)Expression Tree

12. i) Using Dijkstra's algorithm find the length of the shortest path from source A to F. 7



ii) Write the algorithm of insertion sort to arrange some elements in ascending order. Mention best case and average case complexity of this algorithm 5+1

iii). Define complete graph with an example 2

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JGEC/B.TECH/ CSE/ PCC-CS301/ 2025-26
2025
COMPUTER ORGANIZATION

Full Marks: 70 Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions 5x2=10

1. What is overflow in two's complement addition and how is it detected?
2. What is spatial locality? Give example.
3. Compute the range of values that can be represented in 2's complement form when 5 bits are allotted.
4. What is the meaning of EAX, AH and AL in intel terminology?
5. Write an expression for average memory access time in a single level cache memory system.

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions 4x15 = 60

6. i) Perform (-17) – (4) using 2's complement method. 4
   ii) In non- restoring method of division there may require a single restoring step at the end to get the final remainder. Give some numerical examples where this situation will arise and why? 4
   iii) Explain the steps in non-restoring method of division when 31 is divided by 5. 7

7. i) When all bits of the biased exponent are zero then value of unbiased exponent is (1-Bias) instead of (0-Bias). Justify the logic. 4
   ii) Is Booth's algorithm faster than conventional multiplication algorithm. Justify your answer with proper examples. 4
   iii) Multiply multiplicand $(-13)_{10}$ by multiplier $(+5)_{10}$ using Booth's algorithm clearly explaining each step. 7

8. i) a) Convert $(-0.0625)_{10}$ into IEEE single precision format. 4
   b) Convert the IEEE single precision floating point number 1000 0000 0101 1000 0000 0000 0000 0000 into equivalent decimal number. 4
   ii) Consider a 9 bit floating point representation based on IEEE floating point format with 1 sign bit, 4 exponent bits, and 4 fractional bits. Show the representable range of normalized and de-normalized numbers (also express each binary value in equivalent decimal value). 7

9. i) Suppose a computer has 64 bit data bus from memory to the CPU. Explain how a low-order interleaved memory system would be 8-times faster than a corresponding high-order interleaved system when that memory is built with 64 chips. 5
   ii) Build a 256MB RAM using only 32MB basic chips using HIGH-order memory interleaving. Draw the circuit diagram. 5
   iii) Given a computing system with two levels of cache (L1 and L2) and a main memory. The first level (L1) cache access time is 2 nanosecond (ns) and the "hit rate" for L1 cache is 95% while the processor is accessing the data from L1 cache. Whereas, for the second level (L2) cache, the "hit rate" is 90% and the "miss penalty" for transferring data from L2 cache to L1 cache is 18 ns. The "miss penalty" for the data to be transferred from main memory to L2 cache is 120 ns. Compute the average memory access time in this system in nanoseconds. 5

10. i) Explain little-endian and big-endian system with examples. 3

ii) Draw the internal organization of a processor containing PC, MAR, MDR, IR, ALU etc. and describe how data flows from one component to another. 6

iii) Describe register, register indirect, base with displacement and base with index addressing modes the Intel IA-32 architecture follows with suitable examples. 6

11. i) Describe how number of comparators differ between direct memory cache and fully associative cache memory system. 3

ii) Let Tag field size is 12 bits in a 2 way set associative mapped cache memory. How will the tag field size and set number field size (in bits) be changed when the system is upgraded to 4 way and 8 way set associative mapped cache respectively (while keeping the cache memory capacity and block size unchanged). 7

iii) Consider a direct mapped memory system with 4M bytes of main memory and 32K bytes of cache memory. The cache line size is 16 bytes and memory is byte addressable. Determine the parameters tag, line (block) number, byte offset. 5

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JGEC/B.TECH/ CSE/ BS-CH301/ 2025-26

BIOLOGY FOR ENGINEERS

Full Marks: 70 Times: 3 Hours

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GROUP-A

[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions [5X2=10]

1. Write two examples of compound proteins.
2. What do you mean by allele?
3. What do you mean by phenotype?
4. What is a zwitterion?
5. What is Vibrio?

GROUP-B

[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions [15X4=60]

6. Describe the ultramicroscopic structure of a bacterial cell with a suitable diagram.
7. Describe the dihybrid cross experiment of Mendel. Mention the 7 different contrasting characteristics found by Mendel in pea plants. (10+5=15)
8. Classify amino acids with suitable examples. (15)
9. Write about the classification of the enzyme. Write about the different theories of enzyme action. (10+5=15)
10. Briefly describe the classification of bacteria. Describe the various types of sterilisation techniques. (10+5=15)
11. Describe the double helix stranded model of DNA with a suitable diagram. How is a glycosidic bond formed? (10+5=15)

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JGEC/B.TECH/ CSE/HSMC- 301/ 2025-26

ECONOMICS FOR ENGINEERS

Full Marks: 70 Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions 5×2=10

1. Define the term 'unavoidable cost.' 2
2. What is capital expenditure? 2
3. Explain the term 'bills receivable.' 2
4. What is the quick ratio? 2
5. Explain the term 'depreciation.' 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions 4×15 =60

6. What are the objectives of accountancy? Write functions and limitations of accounts. What is engineering economics? Write the role of engineers in the economic decision-making process. 3+3+3+6

7. What are the problems that arise in the economic decision-making process? Write the steps in the decision-making process. Discuss the differences between the trading account and the profit and loss account. 4+4+7

8. What is ratio analysis? Write the advantages and limitations of ratio analysis. How the financial position of a business house can be analyzed using financial ratios. What are the factors for determining depreciation? 3+4+4+4

9. Attempt a short note on each of the following terms:
(i) Current Ratio, (ii) Gross Profit Ratio, (iii) Net Profit Ratio, (iv) Opportunity Ratio, (v) Stock Turnover Ratio. 3+3+3+3+3

10. What is inflation? Write the effects of inflation. What are the causes of inflation? What is cash flow? Write the objective of cash flow. 3+3+3+3+3

11. Distinguish between the following terms: (i) Capital Expenditure and Revenue Expenditure, (ii) Manufacturing Account and Trading Account, (iii) Trial Balance and Balance Sheet, (iv) Bin Card and Store Ledger, (v) Manufacturing Account and Trading Account. 3+3+3+3+3

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
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JGEC/B.TECH/CSE/MC 301/ 2025-26

ESSENCE OF TRADITIONAL KNOWLEDGE

Full Marks: 70 | Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions: 5x2=10

1. Which language is the root of all Indian languages? 2
2. Which traditional Indian knowledge system is directly related to architecture and town planning? 2
3. Who is known as the father of Ayurveda? 2
4. Which ancient Indian text is considered the foundation of Yoga philosophy? 2
5. Name the Indian mathematician who is credited with the invention of the concept of Zero. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions: 4x15 = 60

6. Write an essay on the scope and importance of Traditional Knowledge.
7. Examine the role of women, elders, and community leaders in safeguarding and transmitting traditional knowledge.
8. Discuss the historical development and significance of Ayurveda as a traditional system of medicine in India.
9. Evaluate the role of traditional agricultural engineering practices (terrace farming, irrigation channels) in promoting climate-resilient farming.
10. Examine the roles of youth and administrators in promoting the protection of Traditional Knowledge in India.
11. Discuss the importance of Indian traditions in the field of Engineering and Technology.

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JGEC/COE/B.TECH./CSE/ESC301/2025-26

2026

DIGITAL ELECTRONICS

Full Marks: 70

Times: 3 Hours

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Candidates are requested to write their answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions

5x2=10

1. How many don't't care inputs are there in a BCD adder?
2. Convert the binary $[10001101]_2$ to Gray code and Gray Code $[1100111]_G$ to binary form.
3. Write two applications of multiplexer.
4. Can a shift register be used as a counter? Explain.
5. What is meant by modulus of a counter?

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions

4 x 15=60

6. i) State the DeMorgan's theorems and implement them with the help of NAND and NOR logic gates. 2+2

   ii) A combinational logic circuit has 3 inputs A, B, C and output F. F is true for following input combinations: a) A is false, B is true b) A is false, C is true c) A, B, C are true d) A, B, C are false. Now answer the following

   i) Write truth table for F. Use the convention true =1 and false =0 ii) Write the simplified expression for F in SOP and POS form. iii) Draw the circuit with NAND gate. 8

   iii) Design a 4-bit adder circuit using full adder. Now convert it into 4-bit subtractor using four XOR gates. 3

7. i) Why does a sequential logic circuit require memory element? Show that D flip flop does not need any memory element for its normal activity. 2+2

   ii) Draw the circuit of a serial in-parallel out shift register and explain its working. Data 0101 is entered into 4 bit serial in parallel out shift register. Draw a diagram to show the states of registers after 1, 2, 3, 4 clock pulses. 6

   iii) Implement a full adder circuit using decoder circuit and MUX circuit. 5

8. i) What is Karnaugh map? What is its utility? How is it drawn? Explain the procedure for grouping of cells in K-map. 4

   ii) Using Quine McCluskey tabulation method, simplify the following function. 5

   $F(A,B,C,D) = \sum m(0,1,2,4,5,7,8,10,13,14,15)$

   iii) Simplify the following functions together with don't care condition d, express the simplified function in the sum of minterms: (i) $F(A,B,C,D)=\Sigma(0,2,3,8,13,14)$, $d(A,B,C,D)=\Sigma(1,4,10)$; 6

   (ii) $F(A,B,C,D) = \Sigma(1,3,4,7,9,15)$, $d(A,B,C,D)=\Sigma(5,6,12,13)$.

9. i) What is SR latch? Draw its logic circuit with a) NOR Gate b) NAND Gate, and explain its working. 5

   ii) What are the drawbacks of SR flip flop? How can we overcome it using JK flip flop? 3

   iii) What is excitation table? Write the application of excitation table of a flip flop. Convert a D flip flop into JK flip flop. 1+1+5

10. i) Implement the given function with an 8:1 multiplexer considering C as the data line. 5

$$F(A, B, C, D) = \sum(0, 1, 3, 9, 10, 12, 13, 15)$$

ii) A combinational circuit is defined by the following three functions: 5

$$FA(x,y,z) = x'y' + yz'; FB(x,y,z) = xz + y'; FC(x,y,z) = x + x'yz$$

Design the circuit with a decoder and external logic gates.

iii) Design a combinational logic circuit so that output Y is equal to input A when control inputs B and C are same. If B and C are different, Y should be High. 5

11. i) A ripple counter has 4 flip flops. The initial 3 states are to be skipped. Find the modulus. 2

ii) Draw the circuit of four bit Johnson counter and explain its working. Draw the timing diagram of a 4 bit Johnson counter. 7

iii) In a synchronous counter with 4 flip flops, t_{pd} of each flip flop is 40 ns and t_{pd} of AND gate is 20 ns. Find the maximum possible frequency for which the counter can be used. 4

iv) How many logic gates are used in a Mod-32 synchronous counter? 2

12 i) What is decoder? Construct a 5 x 32 decoder with 3x8 decoder and 2 x 4 decoder only. 1+4

ii) Consider the following functions $F1(w,x,y,z)= \Sigma m(1,3,4,5,9,10,11) + d(6,8)$ and $F1(w,x,y,z)= \Sigma m(0,2,4,7,8,15) + d(9,12)$ a) Simplify the functions F1, F2, F1+F2 (OR), and F1.F2 (AND) using K-map. 7

iii) What is logic race? What is meant by Don't care condition? Give example. 3